

Day 9 – Cleaning & Split Review

Data Quality and Class Balance

Based on the data quality check done on Day 8, I found that there are no missing values in the dataset.

Out of 8,512 rows, 8,486 rows have repeated values. Initially, these looked like duplicate rows, but after checking the dataset, I found that they are repeated readings caused by the limited resolution of the receiver ADC. There are only 26 different signal values available. Therefore, these repeated values are valid measurements and were not removed.

The dataset contains:

- Label 0: 5,321 samples (around 62.5%)
- Label 1: 3,191 samples (around 37.5%)

This shows that the dataset is slightly imbalanced, but the imbalance is not very high. So, I decided not to use techniques such as oversampling or SMOTE at this stage. Instead, I will check metrics such as precision, recall, and F1-score along with accuracy while evaluating the models.

I also found that the bit column is exactly the same as the labels column. Since this column directly gives the answer that the model is supposed to predict, using it as an input feature would cause data leakage. Therefore, the bit column will not be used for model training.

Dataset Split Strategy

I decided to split the dataset into:

- 70% Training data
- 15% Validation data
- 15% Test data

A stratified split will be used based on the labels column. This helps maintain approximately the same ratio of Label 0 and Label 1 in all three datasets.

The repeated values are considered valid independent measurements, so no special grouping is required before splitting the data.

Handling Class Imbalance and Outliers

Since the class distribution is around 62% for Label 0 and 38% for Label 1, I decided not to perform oversampling or undersampling for now.

After training the baseline models, I will check whether one class performs significantly worse than the other. If needed, I can consider imbalance-handling methods later.

No outliers were removed from the dataset. The signal values fall into two clearly separated groups:

- Label 0: approximately 0.337 to 0.366
- Label 1: approximately 0.518 to 0.625

All these values appear to be valid physical signal measurements, so removing any of them is not necessary.

Final Cleaning Plan

The following decisions were made:

1. Keep all repeated rows because they are valid ADC readings.
2. Remove the bit column from the input features because it causes data leakage.
3. Do not remove any outliers.
4. Use a stratified 70/15/15 split for training, validation, and testing.
5. Do not perform resampling at this stage. This will be reconsidered later if the models show poor performance for one of the classes.